Lab 2: Docker

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ALEXANDRIA

1. Prepare the Files
Create a new directory and move into it:

```
mkdir iti-lab && cd iti-lab
git clone https://github.com/meldafrawi/basic-flask-app.git .
```

2. Create the Dockerfile
Create a file named Dockerfile.

```
FROM python:3.11-alpine

WORKDIR /app

COPY . .

RUN pip install --no-cache-dir -r requirements.txt

EXPOSE 5000

CMD ["python", "app.py"]
```

3. Build and Run
Execute these commands to build and then run with the 100MB limit and Port 80 mapping:

```
# Build the image
docker build -t iti-flask-lab2 .
```

Run with memory limit and specific port mapping

```
docker run -d \
  --name flask-container \
  --memory="100m" \
  -p 127.0.0.1:80:5000 \
  iti-flask-lab2
```

4. Push to Docker Hub

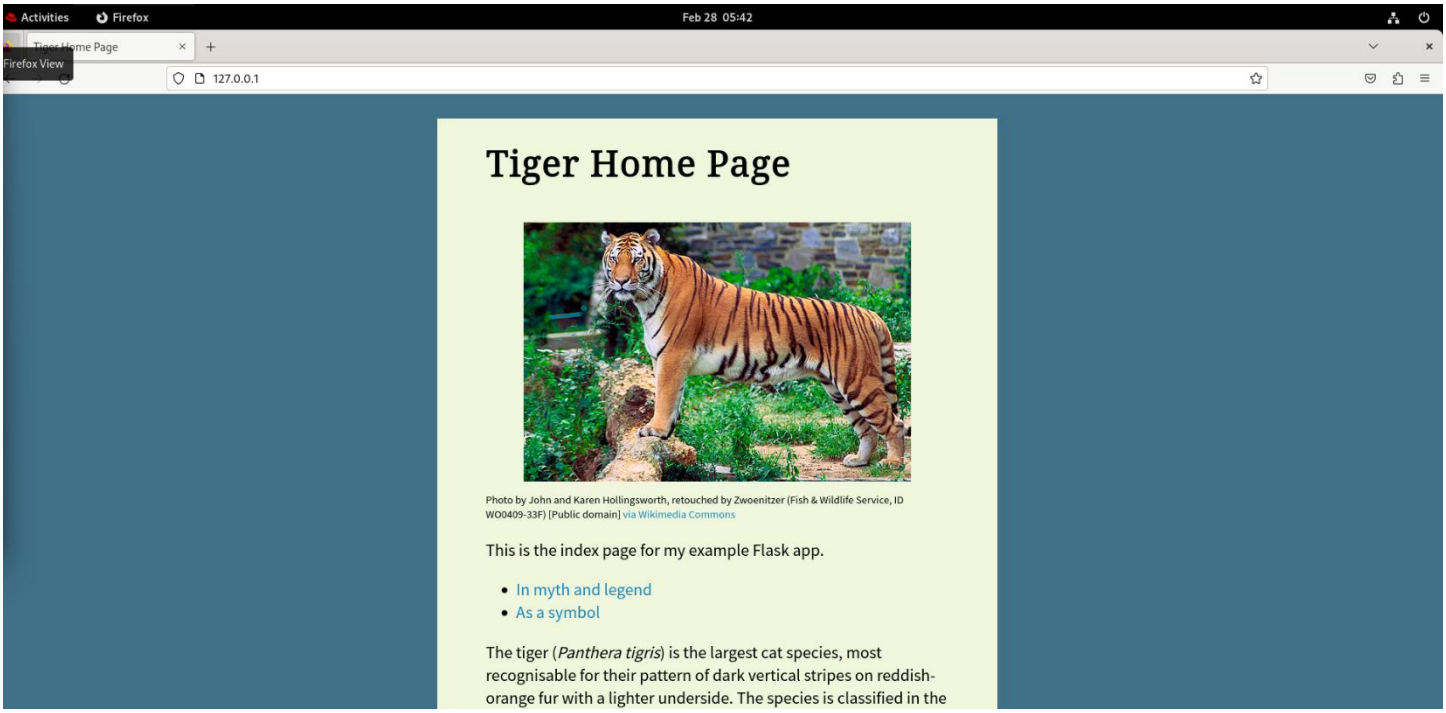
```
docker login

# Replace <your_username> with your actual Docker Hub username

docker tag iti-flask-lab2 oziii/iti-flask-lab2
docker push oziii/iti-flask-lab2
```

to pull it

```
docker pull oziii/iti-flask-lab2
```



Part2 :Network + Nginx:

Create network:

**docker network create --driver bridge --subnet 10.0.0.0/8 \
iti-network**

create dir:

mkdir site

cd site

vim index.html
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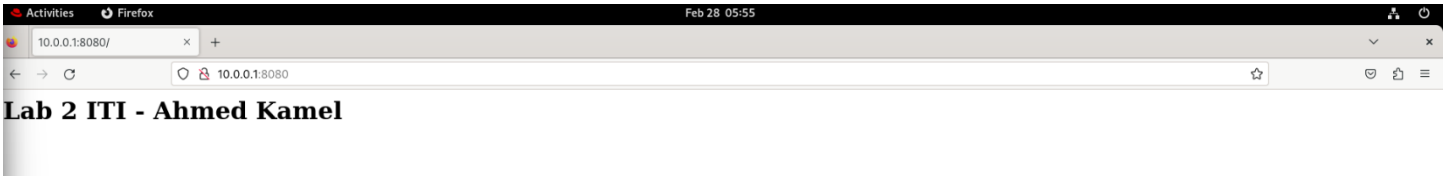
**docker run -d --name webserver-iti \
--network iti-network \
-p 10.0.0.1:8080:8080 \
-v $(pwd)/index.html:/usr/share/nginx/html/index.html:Z \
nginx:alpine**

Because of selinux that nginx work on port 80 not 8080

curl http://10.0.0.1:8080